program01

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1 Introduction to Computer Programming and Artificial Intelligence for Scientists

1.1 Chapter 1: Computers and Programs

Author: Jisu Kim, Otto van Koert, and Joong-Ho Won

2 Textbook

The following book will be the main reference for the first half of the course.

3 Version Info

[ ]:
```
#| label: code-versioninfo
import sys
print(sys.version)
```

---

4 General remarks

- In this semester, Spring 2025, we will conduct the course in English; next year the course will (most probably) be taught in Korean.
- The course is supposed to be very basic (at least at first).
- The main goal is to make everyone familiar with programming and with neural networks (AI).
- Programming will be in python because it is both relatively simple to learn and most AI work is currently being done in python.
- Practice is key for programming, so please do the exercises.

5 The Universal Machine

- A modern computer can be defined as "a machine that stores and manipulates information under the control of a changeable program."
- Two key elements:
  - Computers are devices for manipulating information.
  - Computers operate under the control of a changeable program.

- What is a *computer program*?
  - A detailed, step-by-step set of instructions telling a computer what to do.
  - If we change the program, the computer performs a different set of actions or a different task.
  - The machine stays the same, but the program changes!

- Programs are *executed*, or carried out.
- All computers have in principle the same "power", with suitable programming, i.e., each computer can do the things any other computer can do. (Of course, the execution speed and memory can differ a lot). —

6 Program Power

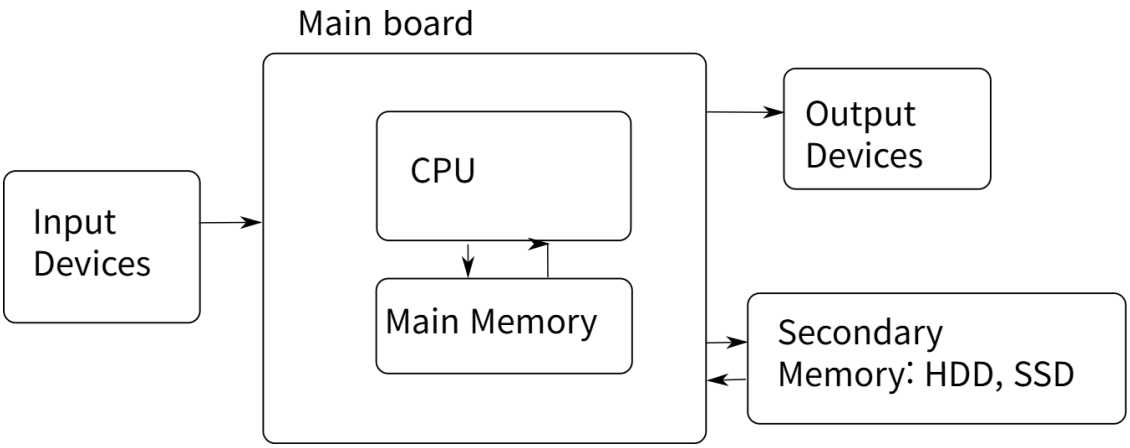
- *Software* (programs) rule the *hardware* (the physical machine).
- The process of creating this software is called *programming*.
- Why learn to program?
  - Fundamental part of computer science
  - Having an understanding of programming helps you have an understanding of the strengths and limitations of computers.
  - Practically speaking, programming is often necessary to automate tasks.
  - Although some programming can be automated using AI, many tasks will require humans to do the real programming work.

7 What is Computer Science?

- Design
  - One way to show a particular problem can be solved is to actually design a solution.
  - This is done by developing an *algorithm*, a step-by-step process for achieving the desired result.

8 Hardware Basics

Here is a photo of the interior of a desktop computer: every computer has most of the visible components.



- The *central processing unit* (CPU) is the "brain" of a computer.
  - The CPU carries out all the basic operations on the data.
  - Examples: simple arithmetic operations, testing to see if two numbers are equal.

- Memory stores programs and data.
  - CPU can only directly access information stored in main memory (RAM or Random Access Memory).
  - Main memory is fast, but volatile, i.e. when the power is interrupted, the contents of memory are lost.
  - Secondary memory provides more permanent storage: magnetic (hard drive, floppy), optical (CD, DVD)
- The displayed computer also has a *graphical processing unit* (GPU) for working with certain repetitive computations, which happen to occur a lot in AI. All the components are connected via the main board or mother board.

- Input devices
  - Information is passed to the computer through keyboards, mice, etc.
- Output devices
  - Processed information is presented to the user through the monitor, printer, etc.

- Fetch-Execute Cycle
  - First instruction retrieved from memory
  - Decode the instruction to see what it represents
  - Appropriate action carried out.
  - Next instruction fetched, decoded, and executed.

You may think of the memory as a long street, where at each address/house there is an instruction or a piece of data. The processor simply retrieves instructions from each address it reaches.

9 Programming Languages

- Natural language has ambiguity and imprecision problems when used to describe complex algorithms.
  - Programs expressed in an unambiguous , precise way using *programming languages*.
  - Every structure in programming language has a precise form, called its *syntax*.
  - Every structure in programming language has a precise meaning, called its *semantics*.

- Programming language like a code for writing the instructions the computer will follow.
  - Programmers will often refer to their program as *computer code*.
  - Process of writing an algorithm in a programming language often called *coding*.

- High-level computer languages
  - Designed to be used and understood by humans
- Low-level language
  - Computer hardware can only understand a very low level language known as *machine language*.

- Add two numbers:
  - Load the number from memory location 2001 into the CPU
  - Load the number from memory location 2002 into the CPU
  - Add the two numbers in the CPU
  - Store the result into location 2003
- In reality, these low-level instructions are represented in binary (1's and 0's)

- High-level language

```
c = a + b
```

- This needs to be translated into machine language that the computer can execute.
- *Compilers* convert programs written in a high-level language into the machine language of some computer.

- *Interpreters* simulate a computer that understands a high-level language.
- The source program is not translated into machine language all at once.
- An interpreter analyzes and executes the source code instruction by instruction.

- Compiling vs. Interpreting
  - Once program is compiled, it can be executed over and over without the source code or compiler. If it is interpreted, the source code and interpreter are needed each time the program runs

  - Compiled programs generally run faster since the translation of the source code happens only once.
  - Interpreted languages are part of a more flexible programming environment since they can be developed and run interactively

10 The Magic of Python

- When you start Python, you will see something like:

```
Python 3.11.4 (main, Jul  5 2023, 09:00:44) [Clang 14.0.6 ] on darwin
Type "help", "copyright", "credits" or "license" for more information.
```

- The "»>" is a Python *prompt* indicating that Python is ready for us to give it a command. These commands are called *statements*.

```
>>> print("Hello, world")
Hello, world
>>> print(2+3)
5
>>> print("2+3=", 2+3)
2+3= 5
>>>
```

- Usually we want to execute several statements together that solve a common problem. One way to do this is to use a *function*.

```
>>> def hello():
        print("Hello")
        print("Computers are Fun")
>>>
```

[ ]:
```python
#| eval: false
def hello():
    print("Hello")
    print("Computers are Fun")
```

- The first line tells Python we are *defining* a new function called hello.
- The following lines are indented to show that they are part of the hello function.
- The blank line (hit enter twice) lets Python know the definition is finished.

[ ]:
```python
#| label: code-hello
def hello():
    print("Hello")
    print("Computers are Fun")
```

- Notice that nothing has happened yet! We've defined the function, but we haven't told Python to perform the function!
- A function is *invoked* or *called* by typing its name.

[ ]:
```
#| label: code-hellorun
hello()
```

- What's the deal with the `()`'s?
- Commands can have changeable parts called *parameters* that are placed between the `()`'s.

[ ]:
```
#| label: code-greet
def greet(person):
    print("Hello",person)
    print ("How are you?")
```

[ ]:
```
#| label: code-greetrun01
greet("Terry")
```

[ ]:
```
#| label: code-greetrun02
greet("Paula")
```

- When we exit the Python prompt, the functions we've defined cease to exist!
- Programs are usually composed of functions, *modules*, or *scripts* that are saved on disk so that they can be used again and again.
- A *module* file is a text file created in text editing software (saved as "plain text") that contains function definitions.
- A *programming environment* is designed to help programmers write programs and usually includes automatic indenting, highlighting, etc.

[ ]:
```
#| label: code-chaos
#| eval: false

# File: chaos.py
# A simple program illustrating chaotic behavior

def main():
    print("This program illustrates a chaotic function")
    x = eval(input("Enter a number between 0 and 1: "))
    for i in range(10):
        x = 3.9 * x * (1 - x)
        print(x)
```

```
main()
```

- We'll use `filename.py` when we save our work to indicate it's a Python program.
- In this code we're defining a new function called **main**.
- The `main()` at the end tells Python to run the code.
- Let's look at the output and then go through this function line by line.

```
>>>
This program illustrates a chaotic function
Enter a number between 0 and 1: .5
0.975
0.0950625
0.335499922266
0.869464925259
0.442633109113
0.962165255337
0.141972779362
0.4750843862
0.972578927537
0.104009713267
>>>
```

11 Inside a Python Program

[ ]:
```
#| eval: false
# File: chaos.py
# A simple program illustrating chaotic behavior
```

- Lines that start with # are called *comments*.
- Intended for human readers and ignored by Python
- Python skips text from `#` to end of line.
- Use `'''` for multiple lines of comments.

[ ]:
```
#| eval: false
print("This program illustrates a chaotic function")
```

- This line causes Python to print a message introducing the program.

[ ]:
```
#| eval: false
x = eval(input("Enter a number between 0 and 1: "))
```

- `x` is an example of a `variable`.
- A variable is used to assign a name to a value so that we can refer to it later.

- The quoted information is displayed, and the number typed in response is stored in `x`.
- In a little more detail, `input` both prints the text and returns the text the user entered. `eval` converts this text into a number.

[ ]:
```
#| eval: false
for i in range(10):
```

- `for` is a *loop* construct.
- A loop tells Python to repeat the same thing over and over.
- In this example, the following code will be repeated 10 times.

[ ]:
```
#| eval: false
x = 3.9 * x * (1 - x)
print(x)
```

- These lines are the *body* of the loop.
- The body of the loop is what gets repeated each time through the loop.
- The body of the loop is identified through indentation.
- The effect of the loop is the same as repeating this two lines 10 times!

[ ]:
```
#| eval: false
for i in range(10):
    x = 3.9 * x * (1 - x)
    print(x)
```

is equivalent to

[ ]:
```
#| eval: false
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
print(x)
x = 3.9 * x * (1 - x)
```

```
print(x)
x = 3.9 * x * (1 - x)
print(x)
```

[ ]:
```
#| eval: false
x = 3.9 * x * (1 - x)
```

- This is called an *assignment* statement
- The part on the right-hand side (RHS) of the "=" is a mathematical expression.
- * is used to indicate multiplication
- Once the value on the RHS is computed, it is stored back into (*assigned*) into x

---

[ ]:
```
#| eval: false
main()
```

- This last line tells Python to *execute* the code in the function `main`.

12 Chaos and Computers

- The chaos.py program:

[ ]:
```
#| eval: false
def main():
    print("This program illustrates a chaotic function")
    x = eval(input("Enter a number between 0 and 1: "))
    for i in range(10):
        x = 3.9 * x * (1 - x)
        print(x)
main()
```

- For any given input, returns 10 seemingly random numbers between 0 and 1
- It appears that the value of `x` is chaotic

- The function computed by program has the general form $cx(1-x)$ where c is 3.9.
- Models certain kinds of unstable electronic circuits.
- Very small differences in initial value can have large differences in the output.

- Input 0.25

```
0.73125
0.76644140625
0.698135010439
0.82189581879
```

```
0.570894019197
0.955398748364
0.166186721954
0.540417912062
0.9686289303
0.118509010176
```

- Input 0.26

```
0.75036
0.73054749456
0.767706625733
0.6954993339
0.825942040734
0.560670965721
0.960644232282
0.147446875935
0.490254549376
0.974629602149
```